

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	xxxxx
Project Title	Smart Lender -Loan Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Smart Lender project aims to simplify the loan approval process using Machine Learning technology. The system predicts whether a loan application is likely to be approved based on the applicant's financial and personal information.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based loan approval prediction system that provides fast and accurate loan eligibility predictions through a simple web application.
Scope	The project focuses on collecting applicant information, preprocessing the data, applying a trained Machine Learning model, and displaying instant loan approval predictions using a Flask-based web interface.
Problem Statement	
Description	Traditional loan approval involves manual verification, making the process time-consuming and prone to human error. Applicants often do not know whether they are eligible before applying.
Impact	Delayed loan decisions reduce customer satisfaction, increase workload for financial institutions, and may lead to inaccurate preliminary assessments.
Proposed Solution	
Approach	Develop a Machine Learning model using Scikit-learn, integrate it with a Flask backend, preprocess applicant information, and generate loan approval predictions through a responsive web application.
Key Features	- Implementation of a ML-based loan approval prediction system. - Real-time prediction using Flask.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 or above GPU
Memory	RAM specifications	8 GB
Storage	Space for dataset, models and project files.	10 GB Free Disk Space
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning Libraries.	scikit-learn, pandas, Numpy,
Development Environment	IDE	Visual Studio Code
Data		
Data	Loan Prediction Dataset	Kaggle Loan Prediction Dataset (CSV Format, 614 Records)